

LAB1 : MongoDB

MongoDB- CRUD Operations Demonstration

```
mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
bmsccse@bmsccse-HP-Pro-Tower-400-G9-PQ-Desktop-PC: ~
bmsccse@bmsccse-HP-Pro-Tower-400-G9-PQ-Desktop-PC: ~
Current MongoDB Log ID: 69f8b697563def5c351e945
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongo-2.4.0
Using MongoDB: 2.4.0
Using MongoSh: 2.4.0
MongoSh 2.8.3 is available for download: https://www.mongodb.com/try/download/shell
For mongoSh info see: https://www.mongodb.com/docs/mongoSh-shell/

-----
The server generated these startup warnings when booting
2026-05-04T13:47:22.247495:36: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2026-05-04T13:47:25.186405:36: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2026-05-04T13:47:25.186405:36: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-05-04T13:47:25.186405:36: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
2026-05-04T13:47:25.186405:36: We suggest setting the contents of sysfsFile to 0.
2026-05-04T13:47:25.186405:36: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
-----

tests> use employee
switched to db employee
employee> db.createCollection('emp_info')
employee> db.emp_info.insertMany([{"id":1,name:"a",desig:"aa",date:ISODate("2026-01-01"),salary:10000,dept:"aaa"},{"id":2,name:"bb",date:ISODate("2026-01-02"),salary:20000,dept:"bbb"},{"id":3,name:"cc",date:ISODate("2026-01-03"),salary:30000,dept:"ccc"}])
{
  acknowledged: true,
  insertedIds: {
    '0' : ObjectId('69f8b6dca7563def5c351e944'),
    '1' : ObjectId('69f8b6dca7563def5c351e945'),
    '2' : ObjectId('69f8b6dca7563def5c351e946')
  }
}
employee> db.emp_info.updateOne({"id":1},{ $set: {name:"David",dept:"ECE"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
employee> db.employee.find().pretty()
employee> db.emp_info.find().pretty()
[
  {
    _id: ObjectId('69f8b6dca7563def5c351e944'),
    id: 1,
    name: 'David',
    desig: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ECE'
  },
]
```



bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~

```
employee> db.emp_info.find().pretty()
[
  {
    _id: ObjectId('69f86dcd7563def5c351e944'),
    id: 1,
    name: 'David',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ECE'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e945'),
    id: 2,
    name: 'b',
    desg: 'bb',
    date: ISODate('2026-01-02T00:00:00.000Z'),
    salary: 20000,
    dept: 'bbb'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e946'),
    id: 3,
    name: 'c',
    desg: 'cc',
    date: ISODate('2026-01-03T00:00:00.000Z'),
    salary: 30000,
    dept: 'ccc'
  }
]
employee> db.emp_info.find().sort({salary:1})
[
  {
    _id: ObjectId('69f86dcd7563def5c351e944'),
    id: 1,
    name: 'David',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ECE'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e945'),
    id: 2,
    name: 'b',
    desg: 'bb',
    date: ISODate('2026-01-02T00:00:00.000Z'),
    salary: 20000,
    dept: 'bbb'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e946'),
    id: 3,
    name: 'c',
    desg: 'cc',
    date: ISODate('2026-01-03T00:00:00.000Z'),
    salary: 30000,
    dept: 'ccc'
  }
]
```



bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~

```
employee> db.emp_info.find().sort({salary:-1})
[
  {
    _id: ObjectId('69f86dcd7563def5c351e946'),
    id: 3,
    name: 'c',
    desg: 'cc',
    date: ISODate('2026-01-03T00:00:00.000Z'),
    salary: 30000,
    dept: 'ccc'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e945'),
    id: 2,
    name: 'b',
    desg: 'bb',
    date: ISODate('2026-01-02T00:00:00.000Z'),
    salary: 20000,
    dept: 'bbb'
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e944'),
    id: 1,
    name: 'David',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ECE'
  }
]
employee> db.emp_info.updateOne({id:1},{ $set:{projects:["data"]}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
employee> db.emp_info.updateOne({id:2},{ $set:{projects:["mach"]}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
employee> db.emp_info.updateOne({id:3},{ $set:{projects:["BDA"]}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```




bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC: ~

```
employee> db.emp_info.find().pretty()
[
  {
    _id: ObjectId('69f86dcd7563def5c351e944'),
    id: 1,
    name: 'David',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ECE',
    projects: [ 'data analytics' ]
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e945'),
    id: 2,
    name: 'b',
    desg: 'bb',
    date: ISODate('2026-01-02T00:00:00.000Z'),
    salary: 20000,
    dept: 'bbb',
    projects: [ 'machine learning' ]
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e946'),
    id: 3,
    name: 'c',
    desg: 'cc',
    date: ISODate('2026-01-03T00:00:00.000Z'),
    salary: 30000,
    dept: 'ccc',
    projects: [ 'BDA' ]
  }
]
employee> db.emp_info.insertOne({id:4,name:"d",dept:"dd",date:
{
  acknowledged: true,
  insertedId: ObjectId('69f874177563def5c351e947')
})
employee> db.emp_info.createIndex({expireAt:1},{expireAfterSe
expireAt_1
employee> db.emp_info.find().pretty()
[
  {
    _id: ObjectId('69f86dcd7563def5c351e944'),
    id: 1,
    name: 'David',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ECE',
    projects: [ 'data analytics' ]
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e945'),
    id: 2,
    name: 'b',
    desg: 'bb',
    date: ISODate('2026-01-02T00:00:00.000Z'),
    salary: 20000,
    dept: 'bbb',
    projects: [ 'machine learning' ]
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e946'),
    id: 3,
    name: 'c',
    desg: 'cc',
    date: ISODate('2026-01-03T00:00:00.000Z'),
    salary: 30000,
    dept: 'ccc',
    projects: [ 'BDA' ]
  },
  {
    _id: ObjectId('69f874177563def5c351e947'),
    id: 4,
    name: 'd',
    desg: 'dd',
    date: ISODate('2026-01-04T00:00:00.000Z'),
    salary: 40000,
    dept: 'ddd',
    projects: [ 'data analytics' ]
  }
]
```



```
}
]
employee> db.emp_info.insertOne({id:4,name:"d",dept:"dd",date:ISODate("2026-01-04T00:00:00.000Z"),
{
  acknowledged: true,
  insertedId: ObjectId('69f874177563def5c351e947')
}
employee> db.emp_info.createIndex({expireAt:1},{expireAfterSeconds:0})
expireAt_1
employee> db.emp_info.find().pretty()
[
  {
    _id: ObjectId('69f86dcd7563def5c351e944'),
    id: 1,
    name: 'David',
    desg: 'aa',
    date: ISODate('2026-01-01T00:00:00.000Z'),
    salary: 10000,
    dept: 'ECE',
    projects: [ 'data analytics' ]
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e945'),
    id: 2,
    name: 'b',
    desg: 'bb',
    date: ISODate('2026-01-02T00:00:00.000Z'),
    salary: 20000,
    dept: 'bbb',
    projects: [ 'machine learning' ]
  },
  {
    _id: ObjectId('69f86dcd7563def5c351e946'),
    id: 3,
    name: 'c',
    desg: 'cc',
    date: ISODate('2026-01-03T00:00:00.000Z'),
    salary: 30000,
    dept: 'ccc',
    projects: [ 'BDA' ]
  },
  {
    _id: ObjectId('69f874177563def5c351e947'),
    id: 4,
    name: 'd',
    dept: 'dd',
    date: ISODate('2026-01-04T00:00:00.000Z'),
    salary: 40000,
    desg: 'ddd',
    expireAt: ISODate('2026-05-04T10:25:42.188Z')
  }
]
employee> 
```